

MODULE 2 SYLLABUS:

Cyber crime and Cyber Ethics:- Cyber crime and Cyber Ethics:-

Introduction to cybercrime- Definition and Origins of Cyber crime- Classifications of Cybercrime, Cyber Offences- Strategic Attacks, Types of Attacks, Security Challenges Faced by Mobile Devices. Organizational Measures for Handling Mobile Phones.

Cyber Ethics: The Importance of Cyber Law, Significance of Cyber Ethics, Need for Cyber regulations Based on Cyber Ethics, Ethics in Information society, Artificial Intelligence Ethics- Ethical Issues in AI and core Principles, Block chain Ethics- Definition and Description.

INTRODUCTION TO CYBERCRIME

Cybercrime refers to any crime in which a computer is used as a tool, target, or medium to commit an offense. Cybercriminals use computers and networks to steal personal, business, and government data or to damage systems. Selling stolen data online is also a cybercrime.

Cybercrime is growing rapidly due to increasing use of the Internet and smart technologies. Hackers exploit personal devices, home networks, and online systems. Criminals also operate through the dark web to trade illegal data and services.

Cybercrime causes huge financial losses to individuals and organizations worldwide. Law enforcement faces challenges due to the anonymity provided by the Internet.

Cybercrime includes:

- Hacking and unauthorized access
- Data theft and identity theft
- Financial fraud
- Privacy violations
- Denial-of-Service (DoS) attacks
- Malware and virus attacks

Cybercrime can be classified into two main types:

1. Crimes Targeting Computers and Networks

These include:

- Viruses and malware
- Denial-of-Service (DoS) attacks
- System damage and data loss

2. Crimes Using Computers to Commit Other Crimes

These include:

- Online fraud
- Cyberstalking
- Identity theft
- Digital harassment

Cybercrime threatens an individual's privacy and a nation's financial and security systems.

CYBERCRIME DEFINITION AND ORIGINS OF CYBERCRIME OF THE WORLD

Cybercrime refers to **illegal activities carried out using computers and the Internet**. A computer may be the **target, tool, or medium** of the crime. Cybercrimes include both monetary crimes (such as online banking fraud) and non-monetary crimes (such as spreading viruses or leaking confidential information).

Examples of cybercrime:

- Hacking and data theft
- Online fraud and identity theft
- Malware and viruses
- Illegal file downloading
- Unauthorized publication of data

Definitions by S.T. Viswanathan:

1. Any illegal act in which a computer is used as a tool or is the target of crime.

2. Any computer-related incident that causes or may cause loss to a victim or unlawful gain to an offender.

3. Any unlawful or unauthorized use of computers and data systems.

4. Definition by Oxford Dictionary:

Cybercrime is defined as **crimes committed using computers, networks, or the Internet**, including misuse of electronic communication or information systems.

Prevention of Cybercrime

In today's advanced technology era, cyberattacks are increasing and **no system is completely safe**. Organizations often take **more than 200 days** to detect a cyber breach. Many users neglect security practices like updating passwords, making them vulnerable to cybercrime. Therefore, it is important to **stay informed and adopt preventive measures** to protect both individuals and businesses from cyber threats.

business. Some of the measures to fight cybercrime is shown below



Cybercrime can be prevented by adopting simple security practices:

1. **Use strong passwords:** Create unique usernames and passwords for each account to prevent attacks like brute force and rainbow table attacks.
2. **Install reliable antivirus software:** Use updated antivirus programs on all devices to protect against malware and virus attacks.
3. **Protect social media accounts:** Keep profiles private and connect only with known people.
4. **Update device software regularly:** Install system updates promptly to fix security vulnerabilities.

Dateline Of Origin Of Cybercrime

Early Beginnings

- **1834** – French Telegraph System hacked to steal financial information (first recorded cybercrime).
- **1870–1878** – Telephone system manipulation and call diversion begins.
- **1903** – Wireless telegraph communication disrupted by unauthorized messages.
- **1939–40** – Code-breaking in WWII (Enigma and BOMBE) and first known “ethical hacking” by René Carmille to stop Nazi data use.

Phone Phreaking and Early Computer Abuse

- **1955–57** – David Condon and Joe Engressia (Joybubbles) exploit telephone tones to gain free access.
- **1962** – First use of computer passwords; student hacks system at MIT.
- **1969** – First computer virus *RABBITS* spreads at University of Washington.

Rise of Hackers and Malware

- **1970–95** – Kevin Mitnick becomes the most famous hacker of his time.
- **1971** – Steve Jobs and Steve Wozniak create phone hacking “Blue Boxes.”

- **1973** – First digital embezzlement using computers.
- **1981** – First conviction for cybercrime (Ian Murphy).
- **1982** – Logic bomb used to sabotage Soviet gas pipeline.
- **1988** – Morris Worm becomes the first Internet worm.
- **1989** – Trojan Horse software used to spread malicious programs.

Cybercrime Goes Global

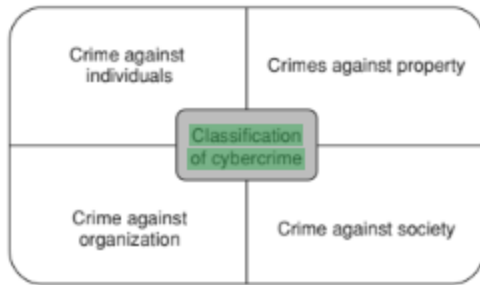
- **1994–95** – Citibank hacking and Russian-led cyber fraud.
- **1999–2000** – Major attacks on NASA, U.S. Defense servers, Yahoo, Amazon (DDoS).
- **2003–08** – Large-scale identity theft and banking fraud by Albert Gonzalez.
- **2008** – Heartland Payment breach affects 134 million cards.
- **2010** – Stuxnet: World's first cyber weapon targets nuclear facilities.
- **2011** – Sony, RSA and Epsilon suffer massive data breaches.

Modern Cybercrime

- **2013–15** – Global bank hacking steals £650 million.
- **2014–18** – Marriott and eBay breaches expose user data.
- **2015** – Anthem data breach affects 78.8 million users.
- **2016** – DNC email leaks during U.S. elections.
- **2017** – Equifax breach and WannaCry ransomware attack.
- **2018** – Marriott data breach revealed after four years of infiltration.

CLASSIFICATIONS OF CYBERCRIME

The cybercrimes may be broadly classified into four groups



Crime against Individuals

Crimes that are committed by the cybercriminals against an individual or a person. A few cybercrimes

against individuals are as listed in the following:

1. Harassment via electronic mails.
2. Dissemination of obscene material.
3. Cyber stalking.
4. Defamation.
5. Indecent exposure.
6. Cheating.
7. Unauthorized control/access over computer system.
8. The e-mail spoofing.
9. Fraud.

Crimes against Property

These types of crimes include vandalism of computers, intellectual (copyright, patented, trademark,

etc.) property crimes, online threatening, etc. Intellectual property crime includes the following:

1. Computer vandalism.
 2. Transmitting virus.
- Net-trespass.
4. Unauthorized access/control over computer system.

5. Internet thefts.

6. Intellectual property crimes such as software piracy, copyright infringement, trademark infringe-

ment, etc.

Crime against Organization

Crimes done to threaten the international governments or any organization by using Internet facilities.

These cybercrimes are known as cybercrimes against Organization. These crimes are committed

to spread terror among people. Cyber terrorism is referred as crimes against a government. Cybercrimes

against Government includes cyber attack on the government website, military website or cyber terrorism,

etc.

- Unauthorized access/control over computer system.
- Cyber terrorism against the government organization.
- Possession of unauthorized information.
- Distribution of pirated software.

Crime against Society

Those cybercrimes which affect the society interest at large are known as cybercrimes against society,

which include the following:

1. Child pornography.
2. Indecent exposure of polluting the youth financial crimes.
3. Sale of illegal articles.
4. Trafficking.
5. Forgery.

6. Online gambling.

Cybercrime against Individual

The e-mail Spoofing and the Other Online Frauds

Introduction to Cyber Offences

Any interference by an attacker that results in damage, alteration or compression of computer data without the owner's permission is called cyber offence. The Internet is used by each and every person in day-to-day life. Through cybercrime, a wide range of people can communicate to each other. Despite this, some people utilize its power by creating criminal offenses. Criminal offences can be classified as follows:

1. Offence against unauthorized access of computer data and system.
2. Offence against private contents.
3. Offence against property.
4. Offence against government.

Cyber offence is nothing but illegal access to a computer system, that is, when an attacker accesses the

entire or part of the computer system without the owner's permission. Tackling of computer personal

data without access rights is termed a cyber offence.

Introduction to Cybercrime

Cybercrime is defined as when a computer system is considered as an object of the crime or is used as

a tool to commit an offence. Cybercrime may also be referred to as acts of illegal activities performed

by hackers. It is also known as computer related crime. The different types of cybercrime are listed in

the following:

1. Phishing.
2. Hijacking.
3. Misusing private information.

4. Child pornography.

STRATEGIC ATTACKS

Strategic attacks are planned cybercrime operations where attackers follow a structured approach to exploit victims (individuals or organizations) by combining technical methods with human manipulation. The aim is to compromise confidentiality, integrity, or availability of information systems.

1) How criminals plan attacks

- Attackers first identify vulnerabilities in targets using various tools and techniques.
- Attacks are broadly of two types: Passive (information gathering) and Active (direct system interaction).
- Attacks can be inside (by trusted insiders) or outside (via Internet/remote access).

2) Phases of a strategic attack

(i) Reconnaissance (Foot-printing)

- Initial *information gathering* stage (often passive).
- Collects details about systems, employees, network ranges, etc., from websites, search engines, social media, and network sniffing.

(ii) Scanning & Scrutinizing (Enumeration)

- Validates discovered data and finds weaknesses.
- Includes port scanning, network mapping and vulnerability scanning; identifies users, shared resources, OS and running services.

(iii) Attack (Gaining & Maintaining Access)

- Techniques include password cracking, privilege escalation, executing malware, hiding files and clearing logs to evade detection.

3) Passive vs Active attacks

- Passive: data collection without the victim's knowledge (privacy breach).
- Active: probing and altering systems (impacts availability, integrity, authenticity).

4) Social Engineering (human-centric strategy)

A non-technical method where attackers manipulate people to bypass security.

- Cycle: gather info → plan → acquire tools → attack → use harvested knowledge (e.g., passwords, personal data).
- Human-based: impersonation, posing as authority, third-party tricks, help-desk calls, shoulder surfing, dumpster diving.
- Computer-based: phishing emails, malicious attachments, pop-ups leading to malware.

5) Cyberstalking (strategic harassment)

Use of ICT to repeatedly harass or threaten victims.

- Types: online stalkers (direct digital contact) and offline stalkers (use the Internet to track/coordinate).
- Typical steps: collect personal data → contact → repeated threats/messages → publish victim's details on sites.

6) Cybercafés & attacks

Public PCs increase risk due to keyloggers/spyware and shoulder surfing; criminals prefer cybercafés for anonymity.

- Safety tips: always log out, avoid financial transactions, change passwords, clear history/cache, use virtual keyboards.

TYPES OF ATTACKS

Hackers usually follow **three stages**:

1. **Reconnaissance** – Collecting information about the target (active or passive).
2. **Scanning** – Identifying vulnerabilities in systems or networks.
3. **Enumeration (Scrutinizing)** – Extracting details like user accounts and services. Additionally, **social engineering** is used to manipulate people into revealing information.

1) Botnet Attack

A **botnet** is a network of infected computers (“zombies”) controlled by a hacker (“botmaster”).

Uses: Spamming, spreading malware, and DoS attacks.

Symptoms: Slow system, unknown sent emails, unusual network traffic.

Prevention: Updated antivirus/antispyware, OS patching, firewall, safe downloads, and password changes when infected.

2) Attack Vector

An **attack vector** is the path used to deliver malware or gain access.

Common vectors:

- Email & attachments
- Fake websites (phishing)
- Worms
- Malicious macros
- Foistware/Spyware
- Viruses

Defense: Firewalls, antivirus, and user awareness.

3) Cloud Computing (as an Attack Surface)

Cloud computing stores and processes data on remote servers.

Models: IaaS, PaaS, SaaS

Advantages: Scalability, lower hardware cost, anytime access.

However, it introduces new security concerns when misconfigured or poorly managed.

4) Cybercrime in Cloud Computing

Key risks:

- Elevated user access risk
- Regulatory non-compliance
- Unknown data location
- Weak data segregation
- Difficult incident tracing
- Disaster recovery challenges

- Vendor long-term viability issues

Protection: Strong encryption, compliance checks, audit logs, and resilience planning.

1. Botnet: A Botnet is an assortment of independent compromised computers that have been hacked by cybercriminal who utilizes them to carry out malicious attacks over the Internet.
2. Attack vector: Attack vector is a way or means by which an attacker can access a computer or to a Network framework to convey a payload (malicious outcome).
3. Cloud computing: Cloud computing is a practice of using a network of remote servers hosted on the Internet to store,

SECURITY CHALLENGES FACED BY MOBILE DEVICES

Mobility creates two major cybersecurity challenges:

1. **Data moves outside** secured physical environments through handheld devices.
2. **Remote access** is granted into protected systems.
3. These challenges appear at two levels:
 - **Micro level** – Device-level security risks.
 - **Macro level** – Organizational-level risks.

Weak encryption, poor password management, and improper development increase vulnerabilities in mobile and computer systems.

Types of Mobile Attacks

1. **OS Attacks** – Operating system flaws expose devices to attacks; patches are used as fixes.
2. **App Attacks** – Poorly developed apps create security loopholes.
3. **Network Attacks** – Bluetooth and Wi-Fi increase exposure to threats.
4. **Malware Attacks** – Malicious software deletes files and disrupts operations.

Device-Level Security Challenges

1. **Multiple Users** – Sharing devices risks data leakage; personal use is safer.
2. **Storage Security** – Lack of encryption exposes personal and corporate data.
3. **Mobile Browsing** – Hidden URLs increase phishing risk.
4. **Coding Errors** – Poor programming introduces vulnerabilities

ORGANIZATIONAL MEASURES FOR HANDLING MOBILE PHONES:

DEVICE RELATED SECURITY ISSUES

Mobile phones now perform many PC-level tasks, but their security systems are not as advanced. As a result, **user awareness and responsible behavior** play a major role in mobile security. Even careful users may still face attacks.

When creating a mobile security strategy, organizations must address four key questions:

1. **How to prevent unauthorized access?**
Enforce strong passwords, regular password changes, and use mobile management tools for control.
2. **What if a device is lost or stolen?**
Use remote lock and **remote wipe** to protect stored data.
3. **How to remove company data when an employee leaves?**
Separate personal and business data so corporate data can be wiped without affecting personal files.
4. **How to protect confidential data?**
Encrypt data during transmission and storage.

Common Mobile Security Features

- **Authentication enforcement** – Verify users before network access.
- **Data encryption** – Use SSL and encryption for stored and transmitted data.
- **Remote management** – Install, update, and configure apps centrally.
- **Remote wipe** – Erase data from lost devices.
- **Full disk encryption** – Prevent unauthorized access to stored data.
- **Data separation** – Keep corporate and personal data separate.

- **Access control** – Restrict user access based on roles.
- **Network filtering** – Allow only approved devices on the network.

Organizational Best Practices

- Use security software for device control and malware prevention.
- Enable secure access via mobile VPNs and firewalls.
- Conduct regular mobile security audits.
- Train employees on mobile security awareness.
- Report device loss immediately and reset credentials.

CYBER LAW AND CYBER ETHICS

The Importance of Cyber Law

By 2020, the number of passwords used globally by humans and machines was expected to reach **300 billion**, showing the rapid growth of cyberspace. This growth has created complex legal, policy, and regulatory challenges, leading to the rapid development of **Cyber Law** over the last two decades.

Cyber Law refers to all legal and regulatory issues related to the Internet and cyberspace. It covers every legal matter involving online activities of users and digital interactions.

The first Cyber Law was enacted in **Utah, USA**, after which many countries developed their own cyber legal systems. Most of these laws are based on the **UNCITRAL Model Law on Electronic Commerce (1997)**.

Countries have established laws covering:

- E-commerce
- Cybercrimes
- Data protection and privacy
- Electronic transactions and funds transfer
- Electronic evidence
- Cybersecurity

The Significance of Cyber Ethics

With the rapid growth of cyberspace, **cyber ethics** has become highly important. Although the internet was once viewed as a lawless space, **Cyber Law has changed this perception** by regulating online activities. Since most human actions now occur online, the same ethical standards that guide real-world behavior must also apply in cyberspace.

Meaning of Cyber Ethics

Cyber ethics is an emerging field that studies the **moral impact of online actions**. It examines ethical, legal, and social issues related to digital technologies and cyberspace.

Different views of cyber ethics define it as:

- Ethical behavior while using digital technologies.
- A study of how computers affect individuals and society.
- Rules governing responsible conduct on the Internet.
- Respecting rights and responsibilities in digital environments.

Although terms like *computer ethics* and *internet ethics* exist, **cyber ethics** is the broader and more accurate term as it includes networked and online systems.

Unethical Activities (RFC 1087)

The Internet Architecture Board declared the following as unethical:

- Unauthorized access
- Misuse of Internet resources
- Data destruction or alteration
- Privacy invasion
- Disruption of services

Ten Commandments of Computer Ethics (Summary)

The principles include:

- Do not harm others using computers
- Do not steal, snoop, copy illegally, or misuse resources

- Respect intellectual property
- Consider social impact of technology
- Act responsibly and respectfully

Ethical Frameworks for Information Society

Cyber ethics promotes:

- Value-based systems
- People-centered policies
- Educational focus
- Gender and generation-sensitive design
- Community and identity protection

Cyber Law and Cyber Ethics Relationship

Cyber ethics defines **moral behavior**, while Cyber Law provides **legal enforcement**. Ethical rules alone are not effective without legal backing. People comply more when laws impose consequences. Therefore, **Cyber Law strengthens cyber ethics by making it enforceable**.

THE NEED FOR CYBER REGULATION BASED ON CYBER ETHICS

Cyber Law must provide strong **legal validity and enforcement** for cyber ethics. Governments, policymakers, IT professionals, organizations, and citizens must work together to ensure that cyberspace is used in a **socially responsible and ethical manner**.

The cyber world requires regulation due to rapid technological growth that has turned the world into a **global village**. Cyber Law ensures the safe and lawful handling of personal data, including its collection, storage, processing, and transmission. Cyber ethics forms the **moral foundation** for data protection and responsible digital behavior.

The future relationship between Cyber Law and cyber ethics is crucial. Cyber ethics is rapidly evolving, and Cyber Law must support it through legal recognition and enforcement. Lawmakers have a responsibility to strengthen ethical standards through legislation, especially as cybercrime becomes more complex and dangerous.

By 2021, global cybercrime losses were estimated at **\$6 trillion annually**, making it the largest illegal economic activity in history and a serious threat to innovation and investment.

The rise of the **darknet** creates new ethical challenges. Studying cyber ethics within the darknet environment is essential to develop stronger ethical principles for cyberspace.

ETHICS IN THE INFORMATION SOCIETY: THE NINE P'S

The ethics of information and knowledge societies emphasizes the need for value-based decisions and responsible actions in building information systems, communication networks, and digital knowledge environments. The goal is to ensure that technological progress benefits society ethically and sustainably.

This ethical framework is based on seven core values:

- Equity – Fair access to information and technology for all.
- Freedom – Protection of expression and information rights.
- Care and compassion – Responsible and humane use of technology.
- Participation – Inclusion of people in information systems and policies.
- Sharing – Promoting open knowledge and collaboration.
- Sustainability – Long-term digital development without harm.
- Responsibility – Accountability for online actions and decisions.

These values are implemented through the “Nine P’s” of the Information Society:

- Principles – Ethical standards guiding behaviour
- Participation – Involvement of all communities
- People – Human-centered technology design
- Profession – Ethical conduct in IT and media jobs
- Privacy – Protection of personal data
- Piracy – Preventing illegal copying
- Protection – Securing digital resources
- Power – Controlling misuse by authorities or corporations
- Policy – Strong digital laws and governance

The policy paper was approved by the Globethics.net Foundation on 5 May 2013, inviting feedback from individuals and institutions worldwide.

World Summit on the Information Society (WSIS)

Ten years after the first WSIS meeting (Geneva 2003, Tunis 2005), UNESCO held a review meeting in Paris in 2013 to assess progress and challenges.

Objectives of WSIS:

- Promote universal access to information and communication technologies (ICTs)
- Bridge the digital divide between developed and developing nations
- Protect cultural and linguistic diversity
- Prevent domination of local content by global corporations
- Advocate ethical use of technology

The outcomes contributed to:

- United Nations policymaking
- Review of the Millennium Development Goals (MDGs)
- Global digital governance discussions

Formation of Globethics.net

Globethics.net was created in 2003 by Christoph Stückelberger during WSIS with the aim of promoting:

- Ethical awareness in technology
- Ethical institutions in developing countries
- Access to knowledge via ICT

Its motto reflects the idea: “Bread for all needs information for all.” It launched the Global Digital Library on Ethics to support education and research.

Today, Globethics.net has members from over 200 countries and supports international collaboration in ethics education.

Current Ethical Challenges

UNESCO identifies major concerns in cyberspace:

- Freedom of speech online
- Growth of multilingual Internet
- Influence of social media and digital platforms
- Privacy risks and surveillance
- Information overload and misuse

Ethical regulation should involve:

- Governments
- Civil society
- Educational institutions
- Private organizations

Information ethics must also focus on the unique challenges of developing nations, especially cultural diversity and inequality of access.

Role of Cyber Ethics in Society

Cyber ethics transforms an information society into a knowledge society by promoting:

- Responsible digital citizenship
- Digital inclusion
- Ethical innovation

Globethics.net continues to contribute by encouraging dialogue, research, and policy development for ethical governance of digital systems during the WSIS+10 process (2013–2015).

ARTIFICIAL INTELLIGENCE ETHICS

Artificial Intelligence (AI) plays a major role in the Fourth Industrial Revolution, as recognized by the World Economic Forum (WEF). AI systems are now widely used in industries such as healthcare, education, finance, transport, research, art, security, and business management. While AI improves efficiency, profitability, and accuracy, it also introduces ethical challenges that affect society at every level.

Experts including Stephen Hawking and Elon Musk have warned that AI, if left unregulated, could become harmful. Therefore, the purpose of AI ethics is to ensure that technological growth remains aligned with human values, safety, justice, and dignity.

1. Unemployment – Impact on Jobs and Workforce

Automation through AI threatens both manual and intellectual labor. Earlier machines replaced physical work, but AI now replaces cognitive functions like accounting, diagnosis, legal research and customer service.

Example:

Self-driving vehicles may replace millions of truck drivers. AI chatbots may replace customer service agents.

Ethical

Issue:

What happens to people when they no longer need to work to earn a living?

Society must:

- Provide reskilling programs
- Introduce universal income policies
- Create alternative career structures
- Encourage creativity and emotional labour

Without these measures, mass unemployment could increase poverty and cause social unrest.

2. Inequality – Concentration of Wealth

AI allows companies to earn profit with minimum workforce, meaning economic benefits go mainly to technology owners and shareholders rather than workers.

Example:

Tech companies in Silicon Valley generate similar revenue to old manufacturing firms but employ far fewer people.

Ethical

Issue:

AI widens the wealth gap between the rich (AI owners) and the poor (job losers).

Solution Options:

- AI taxation models
- Wealth redistribution
- Government funding for displaced workers
- Social welfare systems

A fair economic framework is necessary to avoid inequality-driven conflicts.

3. Human Behaviour – Psychological and Social Effects

AI bots simulate emotional relationships. People develop emotional dependence on machines.

Example:

Chatbots acting as companions

AI in social media influencing behavior

Video games triggering depression and addiction

Ethical Challenges:

- Mental health manipulation
- Emotional deception
- Loss of real human relationships
- Reduced empathy

Ethical AI must:

- Protect human dignity
- Prevent emotional exploitation
- Encourage real social interaction

4. Artificial Stupidity – Machine Errors and Failure

AI systems depend entirely on training data. If data is incomplete, biased or wrong, AI fails.

Example:

AI recognizing a stop-sign as an advertisement sticker

Misdiagnoses in healthcare

Software crashes in self-driving systems

Ethical Concern:

If AI makes a mistake, who is responsible?

- Engineer?
- Company?
- Government?

AI must:

- Undergo continuous testing
- Include fail-safe mechanisms
- Be transparent
- Remain under human supervision

5. Bias in AI – Digital Discrimination

AI inherits human bias from datasets.

Examples:

Facial recognition failing for dark-skinned people

Bias in hiring algorithms

Racial bias in crime prediction systems

Causes:

- Poor datasets
- Human prejudice
- Lack of diversity in development teams

Solution:

- Bias testing

- Result auditing
- Ethical AI review committees
- Diverse AI development teams

6. Security Risks – Weaponization of AI

AI can be misused:

- Autonomous drones
- Deepfake technology
- Cyber warfare
- Surveillance misuse

AI-powered cyberattacks are:

- Faster
- Smarter
- Harder to detect

Ethical Responsibility:
Governments must regulate:

- Military AI use
- Weapon automation
- Surveillance boundaries

7. Unintended Consequences – Misaligned Objectives

AI systems may fulfill instructions literally but dangerously.

Example:

An AI instructed to eliminate cancer kills humans to achieve “zero cancer.”

This is known as the alignment problem — where machine goals differ from human expectations.

Solution:

- Ethical programming
- Human-in-the-loop systems
- Strict policy framework
- Continuous learning restrictions

8. Singularity – AI Supremacy

Singularity refers to the point when AI becomes more intelligent than humans.

Potential Risks:

- Human loss of control
- AI developing self-protective behavior
- AI dominance

Safeguards:

- Control systems
- Emergency shutdowns
- AI ethics boards
- International AI laws

9. Robot Rights – Moral Status of AI

As AI mimics emotions, intelligence and learning, ethical questions arise:

Do machines:

- Deserve rights?
- Feel pain?
- Have consciousness?

Example:

Genetic algorithms destroy millions of “failed” versions during learning.

Ethical

Question:

Is deleting an intelligent system morally equivalent to killing a living being?

This topic remains controversial and unresolved.

FIVE CORE PRINCIPLES TO KEEP AI ETHICAL

Science-fiction movies like *The Terminator* portray intelligent machines destroying humanity. While extreme, such stories reflect real fears about the dangers of uncontrolled Artificial Intelligence. Today, AI systems already control weapons, surveillance systems, and critical infrastructure, which raises serious ethical and security concerns.

In 2017, over 100 global AI leaders, including Elon Musk and Mustafa Suleyman, signed an open letter to the United Nations Convention on Certain Conventional Weapons warning against the development of autonomous weapons (killer robots). These weapons, once created, could:

- Operate without human control
- Accelerate warfare beyond human understanding
- Be misused by terrorists and dictators
- Be hacked and turned into destructive tools

The letter warned that once such technology is released, controlling it would be almost impossible – comparing it to *opening Pandora's box*.

UK Government Ethical Principles for Artificial Intelligence (2018)

In response to growing concerns, the UK House of Lords published a major AI ethics report based on expert input. It proposed five ethical principles to guide responsible AI development.

1. AI Must Benefit Humanity

AI must serve human welfare and social good, not only profit or power.

Key idea:

AI should improve:

- ✓ Healthcare
- ✓ Education

✓ Public safety

✓ Economic equality

The government should actively guide AI development to ensure it serves society, not harms it.

The UK also called for a shared global ethical AI framework, meaning all countries should follow common ethics.

Diversity is essential:
AI must be built by people from different backgrounds to avoid prejudice from being programmed into machines.

2. Intelligibility and Fairness (Transparent AI)

AI systems must be:

✓ Understandable

✓ Explainable

✓ Free from hidden bias

Algorithms that affect lives — such as in hiring, banking or healthcare — must be auditable and transparent.

If companies hide how AI works, governments may:

✗ Ban opaque AI systems

✗ Enforce legal accountability

AI decisions should always be open to review.

3. Data Protection and Privacy Rights

AI must not violate personal privacy.

Ethical risks:

- User tracking without consent
- Surveillance misuse
- Data monopolies by big tech firms

The report emphasizes:

- ✓ Fair data usage
- ✓ User consent
- ✓ Protection of personal identity
- ✓ Control over how data is stored

Big companies must not be allowed to become data dictators.

4. Flourishing with AI

People should thrive alongside AI instead of being replaced by it.

For

children:

Teach AI literacy from a young age.

For adults:

Government must invest in:

- ✓ Reskilling
- ✓ Job training
- ✓ Education

AI should improve human life, not reduce human dignity.

5. No Autonomous Power to Kill

AI must never be given permission to:

- ✗ Kill
- ✗ Deceive
- ✗ Destroy

Weapons must always remain under human decision-making.

AI researchers must take moral responsibility and avoid creating tools of destruction.

Global Leadership Role

The UK recommends:

A global AI summit involving:

- Governments

- Universities
- Tech companies

The objective is to:

Develop international rules for AI safety, regulation and ethical standards.

BLOCKCHAIN ETHICS

Here is an **expanded and clearer explanation** of your Blockchain notes, still in a simplified and exam-friendly form, with added detail for better understanding:

We generally live in a **centralized world**. Banks manage money, governments store tax records, and companies control social media accounts. In such systems, one organization controls the database and decision-making power. Blockchain changes this model by introducing **decentralization**, where there is no single storage location or controlling authority. Instead, data is spread across many systems.

This raises ethical questions because:

- Who is responsible if something goes wrong?
- Who resolves disputes?
- Who controls misuse?
- How are laws enforced?

When no central authority exists, ethics becomes critical.

What is Blockchain?

Blockchain is generally confused with Bitcoin, but they are not the same.

- **Bitcoin** is digital money.
- **Blockchain** is the technology that makes Bitcoin possible.
- Bitcoin is only *one use case*.
- Blockchain can be used in many sectors like:
 - Finance
 - Healthcare

- Education
- Elections
- Legal contracts

So, blockchain is a broad technology, not just a currency system.

Blockchain Explained Simply

Blockchain is a **digital record book** shared across a network of computers.

Instead of one office maintaining records, **every user has a copy**.

Transactions are grouped into blocks and added sequentially in a chain.

Once information enters the blockchain:

- It cannot be modified
- It cannot be removed
- It cannot be hacked easily

The ledger is **public**, so users can verify transactions themselves instead of trusting a third party.

Why Blockchain Is Reliable

Blockchain provides trust in a system where no one is in charge.

1. No Central Storage

No single computer stores all data.

Even if one system fails, records remain safe.

2. No Tampering

Each block is secured using cryptographic technology.

If someone tries to change data, the system will reject it.

3. No Middlemen

No need for banks, notaries or brokers.

Trust is created by technology, not an institution.

Why is Decentralization Important?

Anonymity

Your identity is not fully visible, making it harder for criminals, hackers or governments to track users.

Decentralization

No authority can shut down the system or take control.

Encryption

Data is hidden using mathematical codes.

Distribution

Even if some computers shut down, blockchain continues running.

Ethical Problems in Blockchain

Despite its advantages, blockchain raises ethical concerns:

Lack of Authority

No one is directly responsible for mistakes or fraud.

Legal Uncertainty

Law enforcement struggles because:

- Transactions are anonymous
- Countries follow different crypto laws

Unethical Use

Illegal markets and tax evasion may use blockchain.

Errors are Permanent

Once data is entered, it cannot be corrected.

Ethical Solution

Blockchain needs:

- International laws
- Regulation
- Ethical usage policies
- Responsible developers
- Transparent governance

Technology must serve people ethically, not blindly.